Darmanis 2017 Glioblastoma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 samples intermix per-neighbourhood (iLISI 1.09→2.89 of 4) on one continuous manifold (single kNN component, 0 islands); no per-patient islands.

Gate 2 — module geography → GO (non-branching)

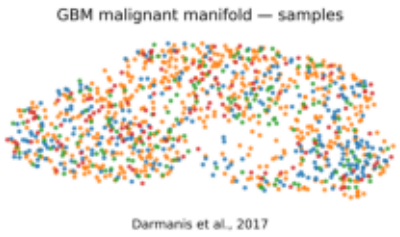
Gast/pSom co-vary (r=+0.66) and both oppose Crest (−0.97 / −0.75): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — hub cross-repression → NA

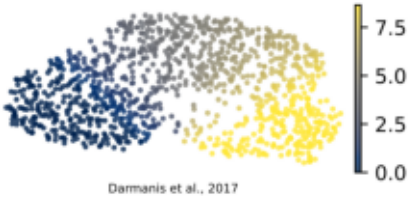
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

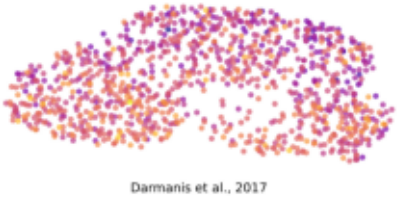
Manifold — samples



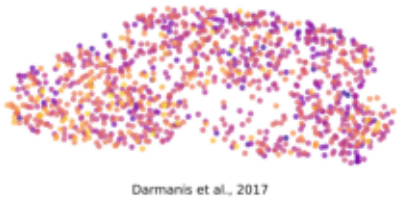
Pseudotime



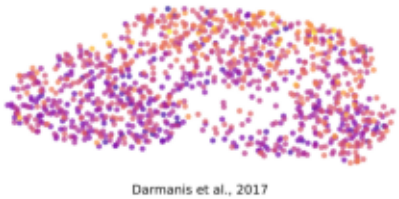
Gastrulation



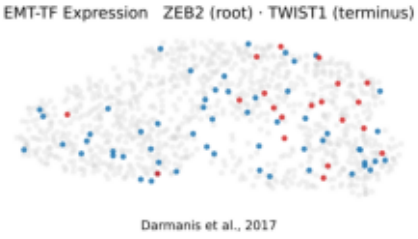
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Darmanis et al., 2017), SmartSeq2 (TPM) — 1,091 malignant cells (4/4 samples).

Verdict.
The embryonic-program pattern REPRODUCES as the non-branching case: a single continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Linear-pSom/Crest geometry (pSom μ=+0.44 strongest). EMT-TFs ZEB2 (root-trending) → TWIST1 (terminus-trending).

Parameter choices

Min-cell threshold	≥100 → 4/4 samples, 1,091 cells (counts 531/268/163/129; ratios <2)	**Expression format**	matrix pre-log-normalized (range 0.7–18.7); undo loader's 2nd log1p (expm1)
Genes / embedding	139 overdispersed (GAM); n_pcs=20; Harmony θ=10	**Islands removed**	0 (single connected kNN component; radius-graph = 1 component)
Geometry	Non-branching — scFates curve N=20 λ=0.3 μ=100; 2 tips, 0 forks, fold-free (2.68)	**Root**	curve tip 0 — Gast/pSom-high, Crest-low origin (Crest late)
EMT-TFs	ZEB2 (51%, root-trending) · TWIST1 (18%, terminus-trending); Snail/FoxC near-absent		